

ALVIN MWANIKI

Python Backend Engineer | Blockchain & Web3 Developer

PERSONAL INFORMATION

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Portfolio: <https://alvin254.netlify.app>

PROFESSIONAL SUMMARY

Backend Engineer with 6+ years of software development experience building and maintaining backend systems, APIs, data-driven applications, and business solutions. Strong expertise in Python, Django, FastAPI, SQL, PostgreSQL, REST APIs, automation, testing, and cloud deployments, with hands-on experience in React, Next.js, and Node.js. Skilled in web scraping and data extraction using Selenium and BeautifulSoup to automate data collection and feed downstream processing pipelines. Recently expanded into blockchain and Web3 development, with practical experience building decentralized applications on the Internet Computer using Motoko, React, and DFX. Experienced in developing reliable software, optimizing systems and data workflows, and integrating third-party services to solve complex business problems.

TECHNICAL SKILLS

- **Programming:** Python, JavaScript, SQL, HTML, CSS,
- **Backend Development:** Django, Flask, Node.js/Express.js, React.js, Odoo, REST APIs, Microservices Architecture
- **Frontend Development:** React.js, Next.js
- **Blockchain & Web3:** Internet Computer (ICP), Motoko, DFX, Decentralized Applications (DApps), DeFi
- **Web Scraping & Automation:** Selenium, BeautifulSoup, Requests, Data Extraction
- **Testing & Quality:** Pytest, Selenium, Unit Testing, Functional Testing, Performance Testing, Test-Driven Development
- **Cloud & DevOps:** AWS (EC2, S3, Elastic Beanstalk), Git, GitHub Actions, CI/CD, Docker, Kubernetes (introductory)
- **Databases:** PostgreSQL, MySQL, SQLite
- **APIs & Security:** API Design and Integration, OAuth2, Third-Party API Integration, Secure API Development

- **Data & Analytics:** Pandas, NumPy, Data Processing, Data Visualization, Dashboard Development
 - **Development Practices:** Agile, Scrum, Code Reviews, Jira, Confluence, Technical Documentation
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PROFESSIONAL EXPERIENCE

Dry Associates

Software Engineer

October 2024 – Present

- Design, develop, and maintain backend systems and microservices using Python and Django, translating business and regulatory requirements into scalable software solutions.
- Develop and integrate REST APIs and third-party services, implementing secure authentication and data exchange mechanisms.
- Build data-driven applications, dashboards, and automated visualizations to support business analysis, reporting, and decision-making.
- Design, optimize, and maintain database solutions using PostgreSQL, MySQL, and SQLite, with a focus on data integrity, performance, and reliability.
- Automate recurring business and operational processes using Python and scheduled jobs, reducing manual intervention and improving workflow efficiency.
- Develop and maintain automated tests using tools such as Pytest and Selenium, covering functional and performance requirements and improving software quality.
- Troubleshoot, debug, and optimize existing applications to improve performance, reliability, and service availability.
- Contribute to CI/CD processes and production releases using Git and GitHub Actions, supporting reliable and repeatable software deployments.
- Participate in system design, code reviews, Agile development practices, sprint planning, and collaboration with stakeholders and cross-functional teams.
- Maintain technical documentation covering system functionality, releases, integrations, and software changes.

IT Support & Development / Engineering

July 2022 – October 2024

- Provided technical support and software development services, troubleshooting hardware, software, network, and application-related issues across the organization.
- Developed and implemented an internal IT ticketing system using Django, streamlining issue reporting and tracking while reducing average issue resolution time by approximately 30%.
- Supported the development, maintenance, and enhancement of internal business applications and software systems.
- Automated routine IT and operational processes using Python and scheduled jobs, improving efficiency and reducing repetitive manual tasks.
- Conducted functional and performance testing, troubleshooting defects and supporting reliable software releases.

- Assisted with database maintenance, data integrity checks, and optimization across PostgreSQL, MySQL, and SQLite environments.
- Integrated third-party APIs and supported secure authentication mechanisms, including OAuth2.
- Developed data visualizations and dashboards to support reporting and operational analysis.
- Supported software deployments, version control, and release activities using Git and GitHub Actions.
- Collaborated with users, developers, and business stakeholders to identify requirements, resolve technical issues, and improve internal systems.
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Web Developer | OwnCloud

April 2021 - July 2022

- Developed and maintained client-facing websites and applications using Odoo, WordPress, and custom solutions, optimizing performance, security, and scalability.
- Built and managed a comprehensive school management system, improving administrative efficiency and user experience.
- Collaborated with multidisciplinary teams to design and deliver innovative technology solutions for diverse business requirements.
- Documented system functionality and performed regular software updates, upgrades,

- and troubleshooting.
 - Supported the integration and optimization of new and existing systems, contributing to overall process and service improvement.
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EDUCATION

Bachelor of Science in Computer Science

St. Paul's University | January 2016 - August 2020

Diploma in Computer Science

St. Paul's University | January 2015 - August 2015

PROFESSIONAL DEVELOPMENT & CERTIFICATIONS

100 Days of Code: The Complete Python Bootcamp

Udemy | March 2023 - January 2024

- **Skills:** NumPy, Pandas, Tkinter, JSON, APIs, Flask, SQLite, PostgreSQL, Web Scraping, Data Visualization
- **Repository:** <https://github.com/nipsalvin/udemy-100DaysOfCode>
- **Certificate:** <https://www.udemy.com/certificate/UC-b02f91c6-edd5-4153-910b-cdb4567b3842/>

Python Django - The Practical Guide

Udemy | February 2024 - August 2024

- **Skills:** Advanced Django Framework Development, AWS S3, EC2, Elastic Beanstalk
- **Repository:** https://github.com/nipsalvin/Python_Django-The_Practical_Guide
- **Certificate:** <https://www.udemy.com/certificate/UC-24713ce5-1e8e-47f6-9bf9-411164224625/>

Web Development Bootcamp

Udemy | September 2024 - Present

- **Skills:** HTML, CSS, JavaScript, jQuery, Node.js, Express.js, OAuth, React.js, DApps, DeFi, Internet Computer (ICP), Motoko
 - **Repository:** <https://github.com/nipsalvin/Web-Development-Bootcamp>
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INTERESTS

Boxing • Programming • Reading • Learning • Travelling

References available upon request